Module 3: Cloud Computing Deployment Models

Quiz for revising on previous work

Q1.
Cloud computing is often described as the *evolution of traditional IT*. Which of the following best explains this shift?
a) IT systems moved from manual operations to purely automated scripts.
b) Organizations replaced owned infrastructure with on-demand, shared resources available over the internet.
c) IT departments started using laptops instead of desktops.
d) Programming languages evolved to include cloud-specific keywords.

Q2.
The NIST (National Institute of Standards and Technology) definition of cloud computing emphasizes five essential characteristics. Which of the following is NOT one of them?
a) On-demand self-service
b) Broad network access
c) Virtual reality integration
d) Resource pooling

Q3.
In traditional IT, scaling resources was costly and slow. In cloud computing, "elasticity" addresses this challenge by:
a) Allowing automatic and rapid adjustment of resources to match demand.
b) Guaranteeing that servers never fail.
c) Offering unlimited free storage to all users.
d) Ensuring that resources are permanently fixed once allocated.

Q4.
Why is the distinction between cloud computing and traditional outsourcing important in defining the cloud?
a) Outsourcing always includes security, while cloud does not.
b) Outsourcing refers to contract-based service delivery, while cloud requires standardized, on-demand, and elastic characteristics.
c) Cloud is only for individuals, outsourcing is for enterprises.
d) There is no real distinction; the terms are interchangeable.

Q5.
Which of the following scenarios best reflects the strategic advantage of adopting cloud computing?
a) A company delays IT investments until hardware prices drop.
b) A startup quickly scales its services to millions of users without purchasing physical servers.
c) An enterprise limits employees' internet use to reduce network costs.
d) A government agency outsources IT helpdesk support to a third-party vendor.

Continued . . . .

Q6.
Cloud service models differ in their level of abstraction. Which model provides the highest degree of control over infrastructure resources?
a) Software as a Service (SaaS)
b) Platform as a Service (PaaS)
c) Infrastructure as a Service (IaaS)
d) Function as a Service (FaaS)

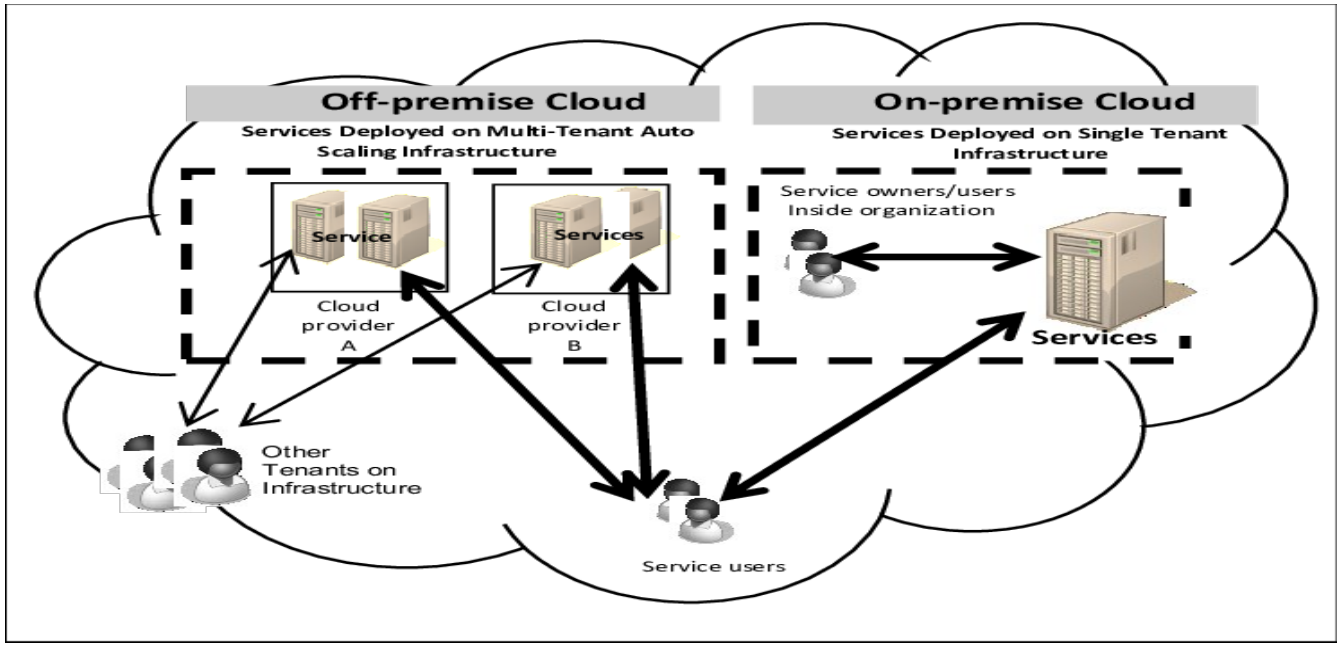
Q7.
One critical enabler of cloud computing is virtualization. Its primary contribution is:
a) Making cloud systems immune to cyberattacks.
b) Allowing multiple virtual machines to share the same physical hardware efficiently.
c) Ensuring applications can run without internet connectivity.
d) Replacing operating systems with browser-based interfaces.

Q8.
Which scenario best demonstrates the "measured service" characteristic of cloud computing?
a) A company is billed monthly based on the exact CPU, storage, and bandwidth it consumed.
b) An employee requests IT support through a ticketing system.
c) A server is manually upgraded to increase processing speed.
d) A business negotiates a fixed price contract for IT services over three years.

Q9.
From a strategic perspective, why do many enterprises adopt a multi-cloud approach?
a) To comply with the NIST cloud definition
b) To reduce dependency on a single provider and optimize cost-performance balance
c) Because single-cloud architectures are illegal in most regions
d) To eliminate the need for disaster recovery planning

Q10.
Which of the following reflects a core offering of leading cloud providers such as AWS, Azure, or Google Cloud?
a) Automated payroll services
b) Infrastructure, platform, and software solutions delivered as services
c) Hardware leasing for on-premises data centers
d) Unlimited free internet access

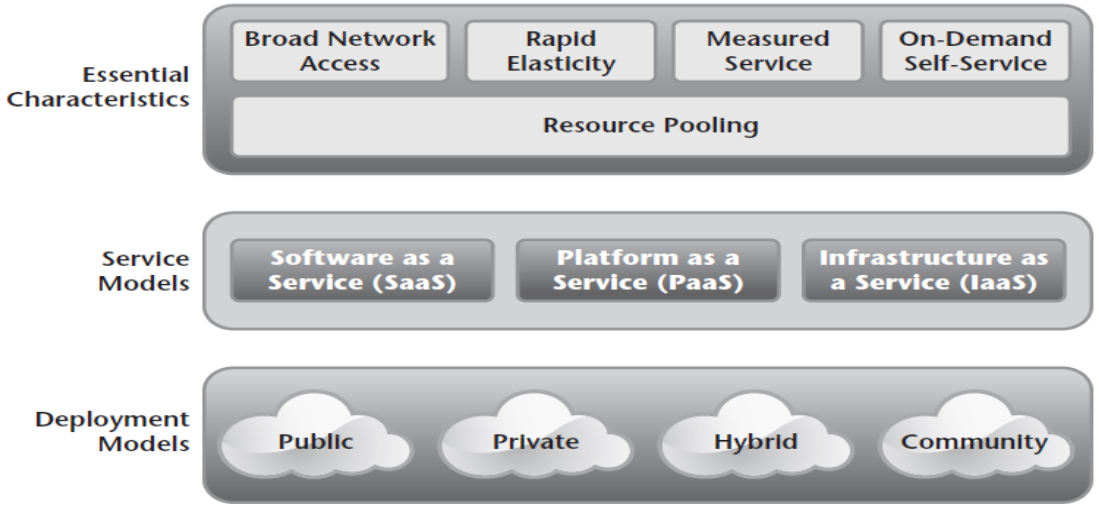
off-premises versus on-premises cloud computing



https://www.researchgate.net/figure/SoA-deployed-on-off-premises-versus-on-premises-cloud-computing_fig1_262374267

Deployment Models

Following Fig. shows the NIST visual model of cloud computing.



Public cloud

A public cloud is a multitenant environment where the end user pays for usage of resources on a shared grid of commodity resources alongside other customers.

The end users have no visibility into the physical location of where their software is running other than where the data centre is located.

Public cloud

- Gmail, Google Drive are examples of public cloud services. Here we are listing various services provided by public cloud companies in the market:

- Microsoft Azure ExpressRoute
- Google Cloud Interconnect
- AWS Direct Connect
- Blue cloud by IBM
- Alibaba Cloud
- Oracle Cloud FastConnect

Benefits of Public Cloud

- Availability
- Scalability
- Accessibility.
- Cost Savings

Drawbacks of Public Cloud

- Integration Limitations:
- Reduced Flexibility
- Forced Downtime

Responsibilities of Public Cloud

- In a public cloud, the service provider is responsible for all the components needed to implement the service.
- The customer is responsible for installing the client-side application and ensuring that it is functioning properly.
- The customer is responsible for general client maintenance.
- The customer is also responsible for providing network connectivity to the provider.

Security Considerations

- Ensuring security is especially difficult in public cloud scenarios.
- Data
- There is also an issue with data access. Theoretically, anyone who works at the service provider could potentially have access to your data.
- Compliance
- Auditing

Private cloud

The cloud infrastructure is provisioned for exclusive use by a single organization comprising multiple consumers (e.g., business units).

The advantage of a private cloud is that it addresses the disadvantages of the public cloud defined earlier

Private cloud end users deploy on a single-tenant environment and are not combined with other customers.

Benefits of Private cloud

- Support and Troubleshooting
- Maintenance
- Monitoring

Drawbacks of Private cloud

- Cost
- Hardware and Software Compatibility
- Expertise Needed

Responsibilities of Private cloud

- In a private cloud environment, the division of responsibilities is pretty straightforward. Your organization will be responsible for the end-to-end solution. You are responsible for the systems that provide the service, the client applications, and the maintenance of the client systems.

Security Considerations of Private cloud

- Ensuring security is easier in a private cloud environment.
- In a private cloud environment, you will be able perform your security and compliance audits.
- Data
- Auditing

Example of Private cloud

- The Cisco Quickstart Private Cloud, which is a self-service private cloud environment
- Dell has a wide range of cloud solutions (hybrid and private) powered by Intel architecture.
- Like Amazon, public cloud giant Google also offers a virtual private cloud (VPC)
- Hewlett Packard Enterprise (HPE) offers an end-to-end private cloud solution portfolio, starting from data centers and infrastructure as a service (IaaS) to consumption services and private cloud consulting.

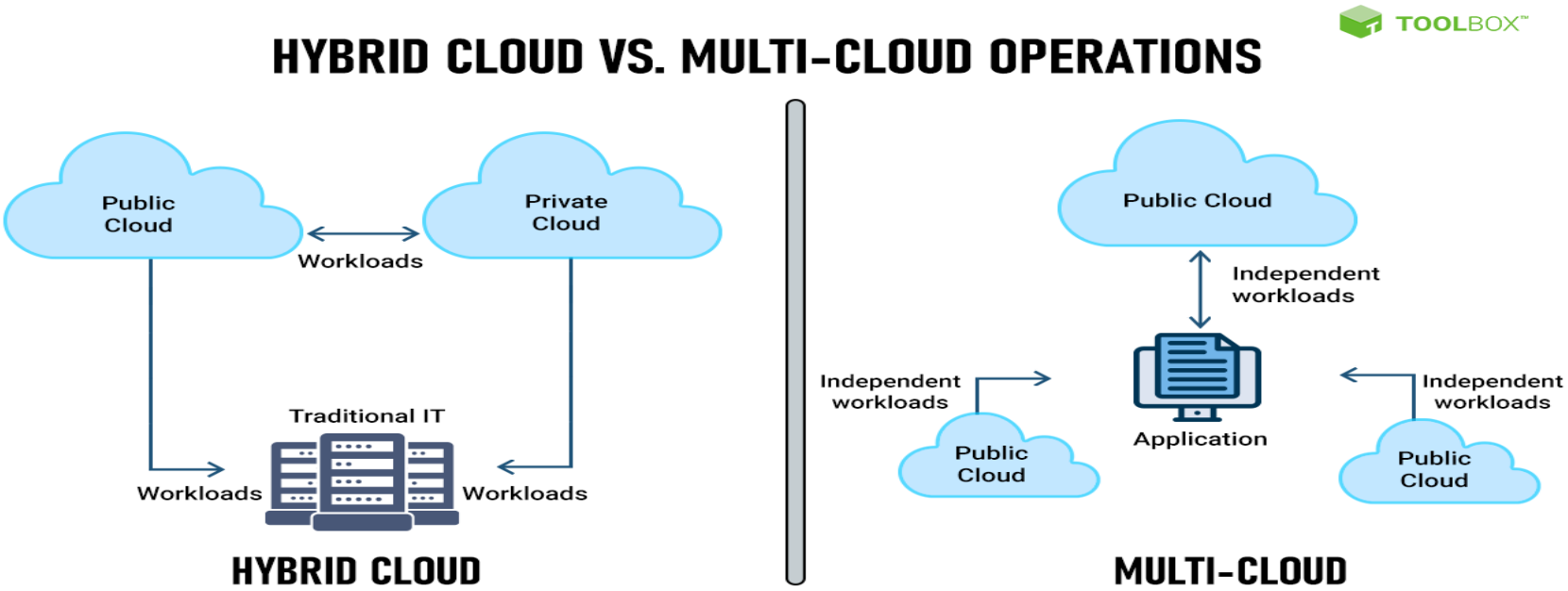
Hybrid cloud

- Hybrid cloud is the combination of compute or storage products from public cloud services (such as Amazon Web Services, Google Cloud Platform, or Microsoft Azure) with a private cloud infrastructure–servers that are generally on-premises running a cloud software stack.

Hybrid cloud

A composition of two or more distinct cloud infrastructures (private, community, or public) that remain unique entities, but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load balancing between clouds).

A best practice for hybrid clouds is to use the public cloud as much as possible to get all the benefits of cloud computing like rapid elasticity and resource pooling, but leverage the private cloud where the risks in areas of data ownership and privacy are too high for the public cloud.



https://www.toolbox.com/tech/cloud/articles/multi-cloud-vs-hybrid-cloud/

Why does hybrid cloud matter?

- Hybrid cloud provides the best of both worlds of computing.
- The Public cloud service providers offer different ability
- Having a private cloud component provides peace of mind.
- Hybrid cloud solutions: IBM, Amazon, Microsoft, AWS Outposts, Azure Stack, Fujitsu Hybrid Cloud Services

Hybrid cloud benefits

- Flexibility
- Security

Certain applications hosted in two environments. That way, if one environment goes down, you can still access the application

Hybrid cloud Drawbacks

- Integration
- Security Considerations
- Data
- Auditing

COMMUNITY CLOUDS

- Community clouds aren't used as much as public or private clouds; in fact, they are the least known and least used cloud deployment model. In a community cloud, the cloud is shared by a group of organizations that have a common purpose or goal.

Benefits:

- Cost
- Multitenancy

COMMUNITY CLOUDS Drawbacks

- Ownership: Ownership in a community cloud needs to be clearly defined. If multiple organizations are coming together to assemble infrastructure, you must determine some agreement for joint ownership.

Cloud Computing Architecture: Roles

End User will consume Cloud service without knowing it's provided by a cloud computing platform. For example, Gmail users.

Cloud Consumer will purchase cloud services to operate their applications. E.g., we can host our website in EC2.

Cloud broker will provide brokering service for cloud computing services, similar to priceline.com for brokering flights, hotels, and car rentals.

Cloud auditor will ensure the cloud is running in a good manner, for example, the federal IT security agency.

Cloud provider is the company that provide cloud computing as a service and charge to the service, such as Amazon.

Cloud carrier is the network operator that help carry cloud service to users.

In class activity : Cloud type

Case Scenario	Cloud Type
A startup hosts its mobile application and databases on AWS to ensure scalability and availability.	
A multinational bank operates its core banking system on an in-house data center but uses Microsoft Azure for customer service chatbots.	
A government agency maintains an internal cloud for confidential citizen records.	
A university uses Google Cloud to host its student learning platform and online courses.	
A network of hospitals collaborates on a shared cloud to securely exchange patient health records and treatment research.	
A logistics company uses IBM Cloud to analyze shipment data and optimize delivery routes.	
A global retail chain runs its customer loyalty program on a private cloud but integrates with a public cloud for AI-based recommendations.	
A law enforcement network shares a cloud infrastructure to store case files and forensic data across different agencies.	
A small business stores its files on Dropbox and Google Drive for easy access and collaboration.	
A pharmaceutical company maintains a private cloud for drug research data but uses AWS for marketing and outreach.	
Several regional banks collaborate on a shared cloud platform to monitor fraud patterns and ensure regulatory compliance.	
A city's smart infrastructure runs its IoT-based traffic monitoring system on a government-managed cloud.	
A gaming company deploys multiplayer game servers on Microsoft Azure, allowing global access to players.	
A university research consortium develops an AI-based climate modeling system using a shared cloud infrastructure.	
A national airline uses a private cloud to store passenger details but utilizes a public cloud for flight booking and customer engagement.	
A multinational media company creates a shared cloud with partners to store and distribute high-resolution video content.	
A university allows students to run machine learning models using a mix of on-campus cloud infrastructure and Google Cloud.	
A defense organization develops classified applications on an in-house, restricted-access cloud.	
A group of non-profit organizations working on environmental research shares a cloud to analyze climate data.	

Learning Activity 1:

Do your research on the internet and find a real-life case study for one of the deployment models discussed in this module (public, private, hybrid or community). Write about the case study in your own words in the discussion forum and include details about the scenario, current and future requirements, and why the chosen model is the best option for the company in no more than 300 words.

As part of the exercise, you may want to consider drawing a mind map outlining the various considerations the company may have had to make when migrating to the cloud. You could do this as a hand-drawn image or take a photo then upload it to the discussion forum or draw it in an application like PowerPoint.

Case Study: Airbnb's Migration to Public Cloud

In its early stages, Airbnb faced challenges with its initial cloud service provider, particularly concerning service administration and scalability. To address these issues, Airbnb decided to migrate nearly all of its cloud computing functions to Amazon Web Services (AWS). This move allowed Airbnb to efficiently manage and customize its infrastructure, enabling rapid scaling to meet the demands of its growing user base. The flexibility and extensive services offered by AWS provided Airbnb with the necessary tools to enhance performance and support its expanding operations. Amazon Web Services

Scenario: Airbnb, a platform connecting hosts with travelers, experienced rapid growth, necessitating a robust and scalable infrastructure. The company required a solution that could handle fluctuating workloads, ensure high availability, and support global expansion.

Current and Future Requirements:

- **Scalability:** Ability to accommodate increasing user traffic and data storage needs.
- **Reliability:** Ensure consistent uptime and quick recovery from potential failures.
- **Global Reach:** Support services in multiple regions worldwide.
- **Cost Efficiency:** Optimize operational costs while maintaining performance.

Why Public Cloud (AWS) Was the Best Option: Migrating to AWS's public cloud offered Airbnb several advantages:

- **Elasticity:** AWS's infrastructure allowed Airbnb to scale resources up or down based on demand, ensuring efficient use of resources.
- **Comprehensive Services:** Access to a wide range of services, including computing power, storage solutions, and databases, facilitated rapid development and deployment of new features.
- **Global Infrastructure:** AWS's worldwide data centers enabled Airbnb to provide low-latency experiences to users across the globe.
- **Cost Management:** The pay-as-you-go pricing model helped Airbnb manage costs effectively, aligning expenses with

Learning Activity 2:

There are various business challenges associated with Cloud deployments. ‘Higher costs’ are regarded as one of them. Read the following white paper titled ‘IT@Intel Brief, A Holistic Cloud Approach for Big Savings’ and answer the following questions: What are the business challenges associated with higher cloud costs? What are the strategies adopted by the vendor to reduce costs? Briefly discuss one of the cost reduction strategies that you think can be applicable if you wish to deploy a service in the cloud. Post your answer to the Module 3 discussion forum.

Business Challenges Associated with Higher Cloud Costs:

1.Slower Time-to-Market (TTM): Without a standard approach to cloud solutions, business groups often face delays and inefficiencies in implementing cloud services, resulting in slower project execution.

2.Inefficient Management: The use of disparate cloud accounts leads to inconsistent technology stacks and a lack of integration. This hampers optimization and alignment of workloads for maximum performance.

3.Higher Costs: Different departments or business units using separate cloud accounts prevents cost consolidation. This makes it difficult to negotiate better pricing, leading to missed opportunities for savings.

Strategies Adopted to Reduce Cloud Costs:

1.Centralized Management: Intel centralized its cloud accounts and workloads, enabling them to optimize resources and negotiate better discounts with cloud providers.

2.Reserved Instances (RIs): By reserving cloud capacity for one or three years, Intel achieved significant savings (around 30-35%) compared to on-demand pricing.

3.Run-rate Discounts: Combining workloads across the enterprise helped Intel understand their cloud spending and qualify for larger discounts from cloud providers.

4.Spot Instances with Artificial Intelligence (AI): Intel uses AI to predict the loss of spot instances ahead of time, allowing them to reallocate workloads to other resources. This approach saved Intel 75-80% over on-demand costs.

Applicable Cost Reduction Strategy for Cloud Deployment:

One strategy that can be particularly applicable when deploying a service in the cloud is using **Reserved Instances (RIs)**. This approach allows you to reserve cloud resources for a fixed period (one or three years) at a significantly lower cost than on-demand pricing. For businesses with predictable workloads or services that do not need frequent scaling, RIs offer substantial savings while ensuring resource availability. By committing to reserved resources, you can avoid fluctuating hourly rates and plan your infrastructure costs in a more stable and predictable manner, ultimately improving cost efficiency.